Cletus: Section 1, Day 3

> To copy code: select it here, or click a "Copy on GitHub" link and use the copy button there. Full note online: https://github.com/mangucletus/AI-Engineering-Course/blob/main/section-1/cletus-section1-day3.md

- Date: [YYYY-MM-DD]
- Course: Week 1 Day 3, types of LLMs, what frontier models are good and bad at, and agentic AI
- Lab notebook: none, this was a lecture day. Practice was done in the chat apps.
- Time spent: [e.g. 1h]

Key concepts

- Base model: trained on a huge amount of text to predict the next token. It continues text but isn't good at following instructions.
- Chat (instruct) model: a base model that was fine-tuned on conversations and human feedback (RLHF) so it follows instructions. ChatGPT is built on this kind of model.
- Reasoning model: trained to think through a problem step by step before answering, using hidden "thinking" tokens. Better at maths, logic and code, but slower and more expensive. GPT-5 Thinking and DeepSeek R1 are examples.
- Main frontier models: GPT (OpenAI), Claude (Anthropic), Gemini (Google), Grok (xAI), plus DeepSeek, which is open-weight.
- What they're good at: summarizing and combining information, turning an outline into a full draft, and writing and explaining code.
- Where they fall down:
- Knowledge cutoff, so they don't know recent events unless they can search the web.
- Niche or specialized topics, where answers can be shallow or wrong.
- Hallucinations, where they sound confident but are wrong.
- Counting letters, since they see tokens and not letters.
- Agentic AI: the model makes a plan and carries it out over several steps using tools. Deep Research searches the web and writes a report, Claude Code writes and runs code in the terminal, and ChatGPT's Agent Mode uses a browser to get things done.
- LLM competition game: models play a game against each other, which shows how differently they behave. It's also a preview of getting several LLMs to talk to each other in Week 2.

Code / examples

No code today. These test prompts are good for comparing models. Copy each one into the chat apps.

1. Letters vs tokens. Models see tokens, not letters, so this trips some of them up.

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How many times does the letter "r" appear in "strawberry"?
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2. Knowledge cutoff. Shows whether the model knows recent news or has to search the web.

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What were the biggest AI announcements this week?
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3. Reasoning. Compare a normal chat model with a reasoning model.

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A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?
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4. Hallucination. Does it invent an answer or admit it doesn't know?

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Summarize the 2019 paper "Recursive Banana Networks for Weather Prediction" by Dr. Lena Okafor.
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(The paper in prompt 4 is made up on purpose. The right answer to prompt 3 is 5 cents.)

Exercise / challenge

1. Run the 4 prompts above in 2 or 3 free chat apps (ChatGPT, Claude, Gemini, DeepSeek).
2. Where the app lets you, turn thinking/reasoning on and off and compare.
3. Keep a small table of which model was right, which was fastest and which explained it best.

Questions & blockers

Things to watch out for: - Don't trust answers about recent events or niche facts without checking them. - The strongest model isn't always the right pick. Reasoning models are slower and cost more.

To discuss with my partner

- Which model did best on which prompt? Did we get the same results?
- When is a reasoning model worth the extra time and cost?

Resources

- ChatGPT: https://chatgpt.com
- Claude: https://claude.ai
- Gemini: https://gemini.google.com
- DeepSeek: https://chat.deepseek.com
- Leaderboard for comparing models: https://lmarena.ai